

習題 1-14

$$1) z^5 e^{\frac{1}{z}} = z^5 \sum_{k=0}^{\infty} \frac{1}{z^k (k!)} = \sum_{k=-\infty}^0 \frac{z^{k+5}}{(-k)!} = \left[\sum_{k=-\infty}^{-5} \frac{z^k}{(-k-5)!}, |z| > 0 \right]$$

$$\begin{aligned} (1) \frac{1}{z^2(z-1)} &= \frac{1}{z-1} \cdot \frac{1}{(z-1+1)^2} = \frac{1}{z-1} [1 + (z-1)]^{-2} = \frac{1}{z-1} \sum_{k=0}^{\infty} \frac{-1}{k!} \cdot (-k-1) \cdot (-k+1) \cdots (-2) \cdot (-1) (z-1)^k \\ &= \frac{1}{z-1} \sum_{k=0}^{\infty} \frac{(k+1)! (-1)^k}{k!} (z-1)^k \\ &= \left[\sum_{k=0}^{\infty} (-1)^k (k+1) (z-1)^{k-1} = \sum_{k=1}^{\infty} (-1)^{k+1} (k+2) (z-1)^k \right] \\ &\quad 0 < |z-1| < 1 \end{aligned}$$

(3)

$$(3) \sin \frac{1}{1-z} = \left[\sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} (1-z)^{-2k-1}, 0 < |z-1| < \infty \right]$$

$$(4) \frac{1}{z^2-3z+2} = \frac{1}{(z-1)(z-2)} = \frac{1}{z-2} - \frac{1}{z-1} = \frac{1}{2} \cdot \frac{1}{\frac{z}{2}-1} - \frac{1}{z-1} = \frac{1}{2} - \frac{1}{2} \sum_{k=0}^{\infty} \left(\frac{z}{2}\right)^k - \frac{1}{z-1}$$

$$= -\frac{1}{2} \sum_{k=0}^{\infty} 2^{-k} z^k - z^{-1} \sum_{k=0}^{\infty} z^{-k} = \frac{1}{2} - \frac{1}{2} \sum_{k=0}^{\infty} \left(\frac{z}{2}\right)^k - \frac{1}{z-1}$$

$$= -\sum_{k=0}^{\infty} 2^{-k-1} z^k - \sum_{k=0}^{\infty} z^{-k-1}, \quad 1 < |z| < 2$$

$$\left[= -\sum_{k=-\infty}^{-1} z^k - \sum_{k=0}^{\infty} 2^{-(k+1)} z^k, \quad 1 < |z| < 2 \right]$$

$$\frac{1}{z-2} = \frac{1}{1-\frac{z}{2}} = \frac{1}{z} \cdot \sum_{k=0}^{\infty} 2^k z^{-k} = \sum_{k=0}^{\infty} 2^k z^{-(k+1)} = \sum_{k=-\infty}^{-1} 2^{-k-1} z^k, \quad 2 < |z| < \infty$$

$$\Rightarrow \frac{1}{z^2-3z+2} = \left[\sum_{k=-\infty}^{-1} 2^{-k-1} z^k - \sum_{k=-\infty}^{-1} z^k = \sum_{k=-\infty}^{-1} (2^{-k-1} - 1) z^k, \quad 2 < |z| < \infty \right]$$

(5) $\sin \frac{1}{z}$ 的奇點 $z=0$.

$$\sin \frac{1}{z} = \left[\sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} z^{-(2k+1)}, \quad 0 < |z| < \infty \right]$$

$$(6) \frac{e^z}{z} = \frac{1}{z} \sum_{k=0}^{\infty} \frac{z^k}{k!} = \sum_{k=0}^{\infty} \frac{z^{k-1}}{k!}, \quad 0 < |z| < \infty$$

(17)

$$\frac{1}{z^2(z^2-1)^2} = z^{-2} \left(\sum_{k=0}^{\infty} z^{2k} \right)^2 = z^{-2} \sum_{k=0}^{\infty} \sum_{l=0}^{\infty} z^{2(k+l)} = z^{-2} \sum_{k=0}^{\infty} \sum_{n=0}^k z^{2(k-n)} z^{2n}$$

$$= \left[z^{-2} \sum_{k=0}^{\infty} (k+1) z^{2k} = \sum_{k=0}^{\infty} (k+1) z^{2(k+1)} \right]$$

$$0 < |z| < 1$$

$$\frac{1}{z^2(z^2-1)^2} = z^{-2} \left(\frac{1}{1-\frac{1}{z^2}} \right)^2 = z^{-6} \sum_{k=0}^{\infty} \frac{1}{z^{2k}} = \sum_{k=0}^{\infty} z^{-2(k+3)} = \sum_{k=-3}^{\infty} z^{-2k}, \quad |z| < \infty$$

習題 1.15

證：記 $z = re^{i\theta}$, $0 < r < 1$

$$\sum_{n=0}^{\infty} r^n \cos n\theta = \operatorname{Re} \left[\sum_{n=0}^{\infty} z^n \right] = \operatorname{Re} \left(\frac{1}{1-z} \right) = \operatorname{Re} \left(\frac{1}{1-r\cos\theta - ir\sin\theta} \right) = \operatorname{Re} \left(\frac{1+r\cos\theta + ir\sin\theta}{(1-r\cos\theta)^2 + r^2\sin^2\theta} \right)$$

$$= \frac{1-r\cos\theta}{1-2r\cos\theta+r^2}$$

$$\sum_{n=0}^{\infty} r^n \sin n\theta = \operatorname{Im} \left[\sum_{n=0}^{\infty} z^n \right] = \operatorname{Im} \left(\frac{1+r\cos\theta + ir\sin\theta}{(1-r\cos\theta)^2 + r^2\sin^2\theta} \right) = \frac{r\sin\theta}{1-2r\cos\theta+r^2}$$

習題 1.16

(1) $\frac{1}{z^2-z^2-2+1} = \frac{1}{(z-1)^2(z+1)}$, $z=1$ 時 = 階極點, $z=-1$ 是 - 階極點

(2) $\frac{\ln(1+z)}{z}$, $z=0$ 是零階極點

(3) $\frac{1}{\sin z^2}$, $z = \sqrt{k\pi}$ ($k \in \mathbb{Z}$) 是 = 階極點

(4) $\frac{1}{(1+z^2)(1+e^{i\pi z})}$, $z = (2k+1)\pi i$ ($k \in \mathbb{Z}$) 是奇點

$z = (2k+1)\pi i$ 是奇點

$z = \pm i$ 是 - 階奇點

$z = (2k+1)\pi i$ ($k \in \mathbb{Z} \setminus \{0, 1\}$) 是 - 階奇點

(5)

習題



习题 1.1

(1) $z = -1$ 是奇点, $\text{Res}(f(-1)) = e^{-1}$ (2) $z = \pm ai$ 是奇点, $\text{Res}(f(\pm ai)) = \frac{e^{\mp a}}{\pm 2ai} = \mp \frac{e^{\mp a}}{2a} i$

($\text{Res}(f(ai)) = -\frac{e^{-a}}{2a} i$, $\text{Res}(f(-ai)) = \frac{e^a}{2a} i$)

(3) $z = 0, \pm 1$ 是奇点, $\text{Res}(f(1)) = -\frac{1}{2}$, $\text{Res}(f(-1)) = -\frac{1}{2}$

$\text{Res} f(0) = \frac{1}{2!} \frac{d^2}{dz^2} \left(\frac{1}{1-z^2} \right) \Big|_{z=0} = \frac{d^2}{dz^2} \left(\frac{1}{1-z} + \frac{1}{1+z} \right) \Big|_{z=0} = -2(z-1)^{-3} \Big|_{z=0} + 2(1+z)^{-3} \Big|_{z=0} = 4$

(4) 当 $n \in \mathbb{Z}^+$ 时, $z = -1$ 是奇点 (n 阶极点)

~~$\text{Res}(f(-1)) = \frac{1}{(n-1)!} \frac{d^{n-1}}{dz^{n-1}} (z^{2n}) \Big|_{z=-1} = \frac{(2n)(2n-1)\dots(2n-n+2)}{(n-1)!} z^n \Big|_{z=-1} = \frac{(2n)!}{(n-1)!(n+1)!} z^{n+1} = \frac{(2n)!}{(n-1)!(n+1)!} (-1)^{n+1}$~~

$\text{Res}(f(-1)) = \frac{1}{(n-1)!} \frac{d^{n-1}}{dz^{n-1}} (z^{2n}) \Big|_{z=-1} = \frac{(2n)(2n-1)\dots(2n-n+2)}{(n-1)!} z^n \Big|_{z=-1} = \frac{(2n)!}{(n-1)!(n+1)!} z^{n+1} = \frac{(2n)!}{(n-1)!(n+1)!} (-1)^{n+1}$

~~当 $n \in \mathbb{Z}^-$ 时, $z = 0$ 是奇点 ($2n$ 阶极点)~~

~~记 $n = -k$~~

~~$\text{Res}(f(0)) = \frac{1}{(2k-1)!} \frac{d^{2k-1}}{dz^{2k-1}} (z^{2k}) \Big|_{z=0} = 0$~~

~~(5) $z = \infty$ 也是奇点~~

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$\frac{z^{2n}}{(z+1)^n} = \left(\frac{z^2}{z+1} \right)^n = \left(\frac{1}{\frac{1}{z} + \frac{1}{z^2}} \right)^n = \left(\frac{1}{\frac{1}{z} (1 + \frac{1}{z})} \right)^n$

$\text{Res}(f(\infty)) = \frac{1}{(n-1)!} \frac{d^{n-1}}{dy^{n-1}} \left(\frac{1}{(1+y)^n} \right) \Big|_{y=0} = \frac{1}{(n-1)!} \cdot \frac{(-n)(-n-1)\dots(-n-n+2)}{(1+y)^{n+1}} \Big|_{y=0} = \frac{(2n-2)!}{[(n-1)!]^2} (-1)^{n-1}$

(5) $e^{\frac{1}{1-z}}$, $z = 1$ 是奇点, $\text{Res}[f(1)] = -1$

(6) $\frac{1}{\sin \frac{1}{z}}$, $z = \frac{1}{k\pi}$ ($k \in \mathbb{Z}$) 是奇点
 $z = \frac{1}{k\pi}$ ($k \in \mathbb{Z}$) 是奇点.
 $\text{Res}(f(\frac{1}{k\pi})) = \lim_{z \rightarrow \frac{1}{k\pi}} \frac{1}{\cos \frac{1}{z} \cdot (-\frac{1}{z^2})} = -\frac{1}{(k\pi)^2 \cdot (-1)^k} = \frac{(-1)^{k+1}}{(k\pi)^2}$

~~$z = \infty$ 也是奇点, $\text{Res}(f(\infty)) = 1$~~



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(7) $\frac{1}{z \sin z}$, $z = k\pi$ ($k \in \mathbb{Z}$) 是奇點.

$$\text{當 } z=0 \text{ 時, } \operatorname{Res} f(0) = \left. \frac{d}{dz} \left(\frac{z}{\sin z} \right) \right|_{z=0} = \lim_{z \rightarrow 0} \frac{\frac{z}{\sin z} - 1}{z} = \lim_{z \rightarrow 0} \frac{1 - \cos z}{z^2} = 0$$

$$\text{當 } z=k\pi \text{ 時, } \operatorname{Res} f(z) = \frac{(-1)^k}{-k\pi}$$

(8) $\frac{z^3}{(\cos z - 1)^2} = \frac{z^3}{4 \sin^4 \frac{z}{2}}$, $z = 2k\pi$ 時奇點

$$\text{當 } z=0 \text{ 時, } \operatorname{Res} f(0) = 4$$

$$\text{當 } z=2k\pi \text{ (} k \neq 0 \text{) 時, } \operatorname{Res} f(2k\pi) = \frac{1}{3!} \left. \frac{d^3}{dz^3} \frac{z^3 (z-2k\pi)^4}{4 \sin^4 \frac{z}{2}} \right|_{z=2k\pi} = 4 + 8k^2\pi^2$$



习题 1.18

$$(1) \oint_{\gamma} \frac{dz}{(z^2+1)(z-1)^2} = 2\pi i (\operatorname{Res} f(i) + \operatorname{Res} f(1)) = -\frac{i\pi}{2}$$

$$\operatorname{Res} f(i) = \frac{1}{(i+i)(i-1)^2} = \frac{1}{2i(i-1)^2} = \frac{1}{2i \cdot 2(-i)} = \frac{1}{4}$$

$$\operatorname{Res} f(1) = \frac{d}{dz} \frac{1}{z^2+1} \Big|_{z=1} = -\frac{2z}{(z^2+1)^2} \Big|_{z=1} = -\frac{1}{2}$$

$$(2) \oint_{|z|=2} \frac{z dz}{\frac{1}{2} \sin^2 z} = \oint_{|z|=2} \frac{2z dz}{\cos 2z} = 4\pi i \frac{1}{2} \oint_{|z|=2} \frac{z dz}{\cos z} = \pi i (\operatorname{Res} f(\frac{\pi}{2}) + \operatorname{Res} f(-\frac{\pi}{2}))$$

$$\operatorname{Res} f(\frac{\pi}{2}) = \lim_{z \rightarrow \frac{\pi}{2}} \frac{z(z - \frac{\pi}{2})}{-\sin(z - \frac{\pi}{2})} = -\frac{\pi}{2}, \quad \operatorname{Res} f(-\frac{\pi}{2}) = \lim_{z \rightarrow -\frac{\pi}{2}} \frac{z(z + \frac{\pi}{2})}{\sin(z + \frac{\pi}{2})} = -\frac{\pi}{2}$$

$$\oint_{|z|=2} \frac{z dz}{\frac{1}{2} - \sin^2 z} = -\pi^2 i$$

$$(3) \oint_{|z|=2} \frac{e^{2z}}{(z-1)^2} dz = \oint_{|z|=2} \frac{e^{2z}}{(z-1)^2} dz = 2\pi i (e^{2z})' \Big|_{z=1} = 4\pi i z e^{2z} \Big|_{z=1} = 4\pi e^2 i$$

$$(4) \oint_{|z-i|=1} \frac{e^z - e^{-z}}{e^z + e^{-z}} dz = 2\pi i (\operatorname{Res} f(\frac{\pi i}{2})).$$

$$\operatorname{Res} f(\frac{\pi i}{2}) = \lim_{z \rightarrow \frac{\pi i}{2}} \frac{(e^z - e^{-z})(z - \frac{\pi i}{2})}{e^z + e^{-z}} = 2i \lim_{z \rightarrow \frac{\pi i}{2}} \frac{1}{e^z - e^{-z}} = 1$$

$$\Rightarrow \oint_{|z-i|=1} \frac{e^z - e^{-z}}{e^z + e^{-z}} dz = 2\pi i$$